CLASS 9 — SCIENCE · FORMULA SHEET

Chapter 11: **Patterns in Life: Diversity and Classification** | Quick revision, one page

Classification and nomenclature

- Taxonomic hierarchy (highest to lowest): Kingdom → Phylum (Division in plants) → Class → Order → Family → Genus → Species.
- Species: group of organisms that can interbreed and produce fertile offspring. Biodiversity: variety of living organisms.
- Binomial nomenclature (Linnaeus): genus (capital first letter) + species (small letter), italics or underlined separately, e.g. *Homo sapiens*, *Mangifera indica*.

Five kingdoms

- Monera: bacteria; unicellular, prokaryotic. Protista: unicellular eukaryotes (*Amoeba*, *Paramecium*).
- Fungi: heterotrophic, chitin cell wall, saprotrophic (mushroom, yeast, bread mould); lichens: association of an alga and a fungus.
- Plantae: multicellular autotrophs with cellulose walls. Animalia: multicellular heterotrophs without cell walls.

Plant groups

- Thallophytes (algae): no differentiated body. Bryophytes: mosses, need water for fertilisation. Pteridophytes: ferns, vascular, no seeds.
- Gymnosperms: naked seeds (pine, cycas). Angiosperms: flowering plants, seeds in fruits; monocots (one cotyledon) and dicots (two cotyledons).

Animal phyla

- Porifera (sponges, pores), Cnidaria (*Hydra*, stinging cells), Platyhelminthes (flatworms, tapeworm), Nematoda (roundworms).
- Annelida (earthworm, segmented), Arthropoda (jointed legs, insects, spiders, crabs; the largest phylum), Mollusca (snail, octopus), Echinodermata (starfish, spiny skin).
- Chordata: notochord and backbone.

Vertebrate classes

- Pisces: gills, two-chambered heart, cold-blooded. Amphibia: moist skin, three-chambered heart. Reptilia: scales, lay eggs on land, three-chambered heart (crocodile has four).
- Aves: feathers, wings, warm-blooded, four-chambered heart. Mammalia: hair, mammary glands, mostly give birth to young, four-chambered heart, warm-blooded.

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